COURSE TITLE : FUNDAMENTALS OF POLYMER SCIENCE
COURSE CODE : 2091
COURSE CATEGORY : B
PERIODS/WEEK : 4
PERIODS/SEMESTER: 60/2
CREDITS : 4

TIME SCHEDULE

Module	Topic	Periods
1	History, General characteristic and classification of polymers	20
2	Chemistry of polymerization and polymerization techniques	15
3	Structure properties and applications of individual polymers	17
4	Polymer degradation	8
TOTAL		60

COURSE OBJECTIVE

Sl.	GO	Student will be able to
1	1	To understand the history, general characteristic ,classification of polymers
2	2	To understand the chemistry of polymerization and polymerization techniques
3	3	To understand structure properties and applications of individual polymers
4	4	To know the polymer degradation

SPECIFIC OUTCOME

MODULE I HISTORY, GENERAL CHARACTERISTIC AND CLASSIFICATION OF POLYMERS

1.1.0 To understand the history, general characteristics and classification of polymers

1.1.1. Summarize the history of polymers

1.1.2. Identify the special characteristics and general application of polymers

1.1.3. Define the terms macro molecules, micromolecules, monomers, oligomers, polymers, degree of polymers, repeating units and functionality with examples

1.1.4. Designate the functionality of monomers with unsaturation and reactive functional groups

1.1.5. Relate the term degree of polymerisation with molecular weight of monomer and molecular weight of polymer and solve with problems.

1.1.6. Classify Polymers with respect to their source, thermal response, application, chemical composition and combination.

1.1.7. Discriminate homo and co polymers, Linear, branched, cross linked, block, graft, random, and alternating co polymers with examples.

1.1.8. Describe the effect of mono, di, and poly functional monomers in structure of polymers

MODULE II CHEMISTRY OF POLYMERIZATION AND POLYMERIZATION TECHNIQUES

2.2.0 To understand the chemistry of polymerization and polymerization techniques.

2.2.1. Define polymerisation.
2.2.2. Classify the polymerisation reaction with typical examples
2.2.3. Define initiators, inhibitor, catalysts, short stops and chain transfer agents with examples.
2.2.4. Explain the mechanism of chain polymerisation viz: free radical ,Ionic and co-ordination with reference to initiation, propagation and termination
2.2.5. Explain step growth polymerisation of polyesters and polyamide.
2.2.6. Explain ring opening polymerisation of caprolactam .
2.2.7. Explain interfacial polymerisation with example.
2.2.8. Explain the polymerisation techniques viz ; Bulk, Solution ,Suspension and Emulsion .
2.2.9 Compare the advantages and disadvantages of different polymerisation technique.

MODULE III STRUCTURE PROPERTIES AND APPLICATIONS OF INDIVIDUAL POLYMERS

3.3 .0 To understand structure properties and applications of individual polymers

3.3.1. Identify the names of General purpose and special synthetic rubbers ,Thermoplastics and Thermoset plastics
3.3.2. State the structure, raw polymer properties and applications of
IR, SBR, BR, IIR, EPDM, NBR, CR, CSM and FKM
3.3.3. State the structure , raw polymer properties and applications of
PE, PP, PVC, PS, PMMA, NYLONS, PET, EPOXY , PC and Polyester
3.3.4. Identify the fibre forming polymers with their structure, properties and applications

MODULE IV POLYMER DEGRADATION

4.4.0 To know the polymer degradation.

4.4.1.Define polymer degradation
4.4.2. Explain chain and random degradation.
4.4.3. List the different causes of polymer degradation.
4.4.4. Describe thermal degradation and mention the factors affecting thermal stability of polymers.
4.4.5. Explain the mechanical and oxidative degradation of polymers.
4.4.6. Illustrate photo degradation of polymers.
4.4.7. Explain the degradation of polymers by hydrolytic, UV rays and ultrasonic waves.
4.4.8. State the advantages and disadvantages of polymer degradation.
4.4.9. Identify the anti degradants and stabilisers used in polymers.
4.4.10.Explain the mechanism of stabilisation of polymers by antidegradants

CONTENT DETAILS

MODULE-I

History , General characteristics and classification of polymer science

History of Polymers, Characteristics and application of polymers –micromolecules ,macromolecules, monomers, oligomers, polymers ,repeating units – examples

Functionality –.-Mono functionality di functionality and poly functionality –examples. Effect of functionality in polymer formation.

Degree of polymerization- molecular weight of polymers –molecular weight of monomers –simple problems.

Relationship between DP and Mw. Classification of polymers –biological-non biological –Natural , synthetic and derived polymers .-Thermo plastic –thermo set polymers –Crystalline and amorphous –organic and inorganic polymers –homo chain and hetero chain polymers-homo and co- polymers –linear branched and cross linked polymers .Alternating and random co -polymers .Examples of each .

MODULE-II

Polymerization reaction and techniques

Type of polymerization reaction –chain and step polymerization with examples –steps in chain polymerization –initiation, propagation and termination. Mechanism of free radical , ionic and co-ordination polymerization. Living polymers ,initiators and catalyst, inhibitor sand chain transfer agents with examples . Polymerisation techniques –bulk solution ,suspension and emulsion polymerization -Comparison

MODULE-III

Structure properties and application of individual polymers Elastomers –General purpose ,special purpose –examples –application Plastics –thermo plastics ,thermo sets –examples –application oF ibers –natural and synthetics –examples .Structure –polymer properties—application of polymers –SBR, IIR, IR, EPDM, BR, NBR, CR, CSM and FKM . PE,PP,PVC,PS,PMMA,NYLONS,PET,EPOXY,PC AND POLYESTER

MODULE-IV

POLYMER DEGRADATION Degradation –chain and random degradation –factors of polymer degradation –Thermal degradation –stability of polymers ,factors affecting thermal stability of polymers –mechanical degradation –oxidative degradation –Photo degradation –hydraulitic degradation-UV rays and Ultrasonic degradation. Advantages and disadvantages of polymer degradation .Anti degradants –stabilizers –anti oxidants anti ozonants .

REFERENCE BOOKS

1. Polymer science -V.R.Gowarikar (Publisher – New Age International (P) Ltd, New Delhi- 2005 Edition
2. Text Book of Polymer Science -Billmeyer.J.R (John Wiley & Sons (Aia) PTE Ltd, Singapore, 2008 Edition
3. Introductory Polymer Chemistry – G.S.Misra (Publisher – New Age International (P) Ltd, New Delhi- 2005 Edition
4. Principles of Polymer Science -P.Bahadur, N.V.Sastri (Publisher – Alpha Science